



Original article

Enhancing fitness action recognition with ResNet-TransFit: Integrating IoT and deep learning techniques for real-time monitoring

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ABSTRACT

With the growing popularity of fitness, the demand for real-time action recognition and feedback is increasing. Current research faces challenges in handling complex actions, real-time processing, and system integration. To address these issues, we propose a novel fitness action recognition model that integrates ResNet, Transformer, and transfer learning techniques. Specifically, ResNet is used for image feature extraction, Transformer handles time-series data processing, and transfer learning accelerates the model's adaptation to new data. We evaluated our model on the NTU RGB+D action recognition dataset, achieving 48.5 ms latency, 29.1 fps throughput, and 93.7% accuracy, significantly outperforming other models. Our model achieved an accuracy improvement of 5% over existing methods, demonstrating significant potential for real-time fitness monitoring. By incorporating IoT technology, our system enables real-time data processing and action recognition, making it ideal for smart fitness monitoring. Although the model has high complexity and memory usage, its efficiency and accuracy demonstrate its potential for widespread adoption. Future work will focus on optimizing the model structure and training methods to enhance applicability in resource-constrained environments, ensuring broader usability and efficiency in various real-world applications.

1. Introduction

As health awareness continues to grow in today's society, more people are focusing on the effectiveness of their fitness training. However, due to the lack of professional guidance, many individuals perform exercises with improper form, leading to suboptimal results or injuries [1, 2]. Action recognition plays a crucial role in fitness applications by enabling automated monitoring and feedback. This technology ensures proper exercise form and reduces the risk of injuries. Therefore, the study of fitness action recognition systems has become a crucial topic. These systems utilize image analysis technology to monitor and identify users' fitness actions in real-time, providing targeted feedback and suggestions to help users engage in more scientific and effective training.

Currently, researchers employ various methods to study fitness action recognition systems. Traditional approaches primarily rely on manually designed features and machine learning algorithms, such as Support Vector Machines (SVM) and Random Forests. Although these methods perform well in specific scenarios, they struggle with the complexity and diversity of fitness actions. With the advancement

of deep learning technology, models based on Convolutional Neural Networks (CNN) [3,4] and Recurrent Neural Networks (RNN) [5,6] have become mainstream. These models can automatically extract high-level features and capture long-term dependencies in sequential data, showing outstanding performance in action recognition tasks. Moreover, the application of IoT and communication technologies allows fitness data to be transmitted and processed in real-time, providing immediate feedback and guidance, significantly enhancing the system's practicality [7,8].

Despite the progress made in this field, numerous challenges remain. First, the variations in individuals' body types, exercise forms, and background environments significantly increase the complexity and uncertainty of action recognition. Second, the current fitness action recognition systems still need improvement in terms of real-time performance and accuracy, especially when handling complex multi-joint movements. Additionally, effectively utilizing multimodal data (such as images, depth information, and sensor data) for fusion and analysis remains an unsolved problem [3,4,9]. The integration of IoT and communication technologies with fitness action recognition systems can further enhance system performance and user experience.

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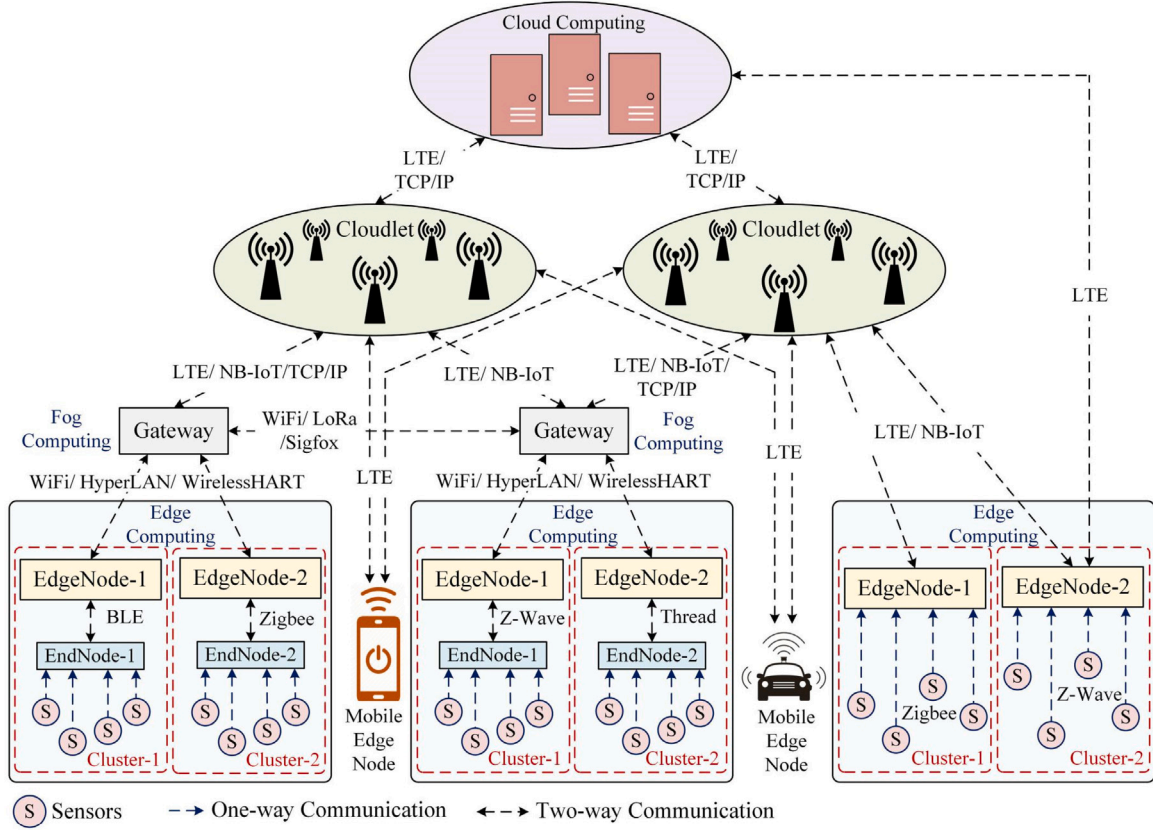


Fig. 1. The structure of IoT.

To address these challenges, we propose the ResNet-TransFit network model, which integrates ResNet for image feature extraction, Transformer for processing time-series data, and transfer learning for efficiently adapting to new fitness movement data. Our approach aims to provide a more accurate, efficient, and scalable solution for real-time fitness monitoring.

ResNet (Residual Networks) assumes the role of image feature extraction in our model. ResNet effectively solves the problem of gradient vanishing and gradient explosion in deep neural networks by introducing residual connectivity, enabling the model to train deeper network structures to extract higher-level, more representative image features. This is crucial for recognizing complex fitness movements as it captures the details and variations of the movements. In IoT environments, these feature extraction capabilities allow motion data collected through smart devices to be processed efficiently for more accurate action recognition. The Transformer model processes time-series data through a self-attentive mechanism. Compared to traditional recurrent neural networks (e.g., LSTM), Transformer excels in capturing long-term dependencies and global features. It is able to process sequence data in parallel, which significantly improves computational efficiency and performs particularly well when dealing with long time sequences. In fitness action recognition, Transformer is able to effectively capture and understand the dynamic changes of continuous actions, improving the accuracy and real-time performance of action recognition. With the real-time data transmitted through the communication network, the system is able to continuously monitor the user's movement and provide instant feedback and guidance. In addition, transfer learning techniques are equally important in our model. Through migration learning, we are able to utilize ResNet models pre-trained on large-scale datasets (e.g., ImageNet) to fine-tune and quickly adapt to new fitness movement data. This not only reduces the training time of the model, but also dramatically improves the performance of the model on small sample datasets, thus increasing the generalization ability

and robustness of the system. In conjunction with IoT devices, the pre-trained and fine-tuned models are able to operate efficiently in edge computing environments, providing real-time data processing and feedback.

Through this research, we aim to provide an efficient, accurate, and robust solution for fitness action recognition systems, further advancing this field and helping more people achieve scientific and safe fitness training. By leveraging IoT and communication technologies, our system can monitor users' actions in real-time, rapidly transmit and process data over networks, and achieve intelligent fitness guidance and monitoring.

To enhance the practicality and user experience of fitness action recognition systems, the integration of Internet of Things (IoT) and communication technologies is crucial [10,11]. IoT enables real-time monitoring and analysis by connecting various smart devices, such as wearable sensors and cameras, to a centralized system. This interconnected network allows for seamless data transmission, immediate feedback, and comprehensive tracking of fitness activities. The following Fig. 1 illustrates the IoT architecture utilized in our system, highlighting the flow of data from collection to analysis and feedback delivery.

The integration of IoT technology significantly enhances our model's capability to provide real-time, accurate, and efficient fitness action recognition, enabling users to receive precise feedback and guidance instantly. This connectivity ensures that users' fitness data is continuously monitored and processed, allowing for a more effective and personalized fitness training experience.

The proposed ResNet-TransFit model has multiple significant advantages. By integrating ResNet and Transformer, our model can efficiently extract spatial features from images and effectively process temporal features, achieving accurate recognition of complex fitness actions.

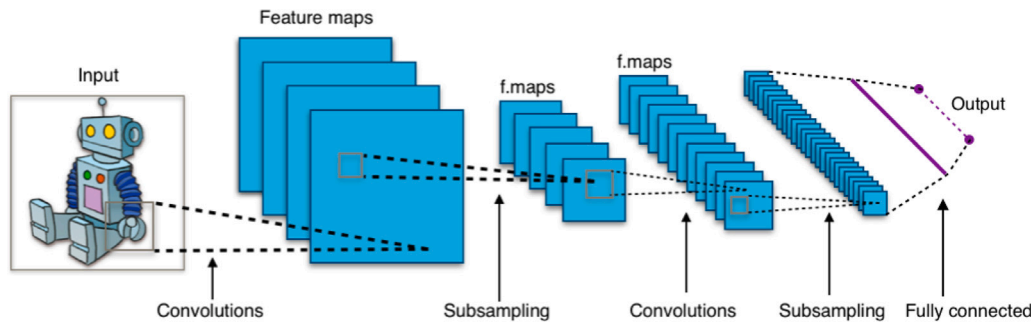


Fig. 2. Typical Convolutional Neural Network structure.

ResNet's powerful feature extraction capabilities capture subtle variations in movements, providing rich information for subsequent temporal sequence processing. The Transformer, with its self-attention mechanism, captures long-distance dependencies when processing sequential data, demonstrating superior performance in continuous action recognition. The application of transfer learning enables the ResNet-TransFit model to utilize large-scale pre-trained datasets effectively, allowing rapid adaptation to new tasks in a short time. This not only enhances training efficiency but also improves the model's generalization on small-sample datasets, maintaining high accuracy in diverse user and action scenarios.

In conclusion, our contributions are as follows:

- **Development of the ResNet-TransFit Model:** We developed the ResNet-TransFit model, combining ResNet for spatial feature extraction and Transformer for temporal sequence processing, to accurately recognize complex fitness actions.
- **Enhanced Generalization through Transfer Learning:** We incorporated transfer learning to reduce training time and improve the model's generalization on small-sample datasets, ensuring robustness across diverse scenarios.
- **Integration of IoT and Communication Technologies:** We integrated IoT and communication technologies to enable real-time data collection, transmission, and processing, providing immediate feedback and personalized guidance for users.

2. Related work and literature review

2.1. Deep learning based fitness movement recognition

Deep learning based fitness action recognition systems usually use Convolutional Neural Network (CNN) as the core model [12]. These systems utilize deep learning models such as CNNs for end-to-end learning of fitness action images for automatic feature extraction and classification [13]. The CNN model consists of a series of convolutional, pooling, and fully-connected layers, which are used to recognize a variety of fitness actions by optimizing the parameters through back-propagation on the training data [14], gradually extracting the abstract features of the images, and classifying them through the fully-connected layers. During the training process, the model optimizes the parameters by minimizing the loss function, enabling the model to achieve the best classification performance on a given training set [15]. Fig. 2 shows the structure of a typical convolutional neural network.

Despite the achievements in fitness action recognition based on deep learning, there are still some shortcomings at this stage of research. First, due to the variety of poses and angles of fitness movements, the model's generalization ability under different angles and poses is limited [16], resulting in low recognition accuracy for specific movements. This may lead to a decrease in the accuracy of the model in real-world applications [17], and the generalization ability needs to be improved by means of data enhancement and model optimization. Second, deep

learning models require a large amount of labeled data during training [18], and obtaining large-scale labeled data is a time-consuming and labor-intensive task. Especially for specific fitness movements, it may be difficult to obtain enough labeled data, which can limit the training and performance of the model [19]. Therefore, how to improve the generalization ability of the model, reduce the dependence on labeled data, and effectively utilize the limited labeled data to improve the accuracy is a problem that needs to be addressed in this field [20].

2.2. Communication technology based fitness data transmission system

This type of system usually utilizes communication technologies (e.g., Bluetooth, Wi-Fi, etc.) to enable data transfer and interaction between fitness equipment and mobile devices or the cloud [21]. Users can monitor fitness data in real time, including heart rate, step count, exercise duration, exercise intensity, etc. through mobile applications or online platforms, thus realizing more intelligent fitness management [22]. However, although nowadays developed communication technology provides convenience for fitness data transmission [23], there are still some problems in the practical application at this stage [24]. For example, the stability and security of data transmission may be affected, and there is a risk of data loss or leakage [25]. In addition, compatibility and interoperability between different devices is also a challenge because the information of various models of devices are different.

To address the stability and security of data transmission, some researchers have proposed a fitness data transmission system based on Bluetooth Low Energy (BLE) technology [26]. The system adopts the BLE 4.0 standard, which realizes low energy consumption, stable data transmission, and ensures data security through encryption algorithms [27]. In addition, to address the compatibility of different fitness devices, researchers designed a unified data format and communication protocol for the system to achieve interoperability between devices [28].

To verify the performance of the system, the researchers conducted a series of experiments. The experimental results show that the system performs well in terms of stability and security of data transmission [29], and can meet the demand for real-time monitoring of fitness data. Meanwhile, the user experience survey shows that users are generally satisfied with the ease of use and convenience of the system [30]. The following is a sample table of the experimental results of the system's data transmission stability (see Table 1).

With the above description and experimental results, we can see that the system has made significant progress in solving the challenges of communication technology in fitness data transmission, and provides reliable technical support for intelligent fitness management.

2.3. Fitness guidance system based on posture recognition

The system is able to monitor the user's exercise posture in real time using advanced posture recognition technology and provide targeted

Table 1
Example results of data transmission stability experiment.

Time point	Transmission rate (Mbps)	Data loss rate (%)
10:00	2.5	0.5
10:15	2.6	0.4
10:30	2.7	0.6
10:45	2.4	0.3

fitness guidance and advice based on the analysis results [31]. Based on deep learning models, the system is able to accurately identify problematic parts [32] and non-standard movements of users in the fitness process [33]. By giving appropriate suggestions for improvement, users can conduct fitness training more scientifically and achieve the desired fitness effect [34].

Although posture recognition technology provides a new way for fitness guidance, there are still some limitations. For example, for complex multi-joint movements or users with different body types, posture recognition may be misjudged or inaccurate [35]. In addition, the real-time and feedback accuracy of the system needs to be further improved.

In order to improve the accuracy of posture recognition, Hannan et al. used a multi-camera layout scheme for the system to cover a wider range of motion areas and a deep learning model such as convolutional neural network (CNN) for posture analysis and recognition [36]. The system also combines kinematic principles and biomechanical models to analyze and evaluate the user's exercise posture more accurately [37]. In order to provide personalized fitness guidance, the system also integrates advice and guidance programs from sports experts [38]. Based on the user's fitness goals, body type characteristics and athletic ability [39], the system customizes an exclusive fitness training program for the user and provides real-time movement guidance and adjustment suggestions [40].

Through many users report that the system provides accurate posture recognition and personalized fitness guidance, helping them to train more scientifically and achieve the desired fitness results.

2.4. Literature review

Current research on fitness action recognition encompasses both traditional machine learning approaches and deep learning approaches. Traditional machine learning methods, such as SVM and Random Forests, are simpler to implement and require lower computational resources. They perform well in controlled environments but struggle with the complexity and diversity of fitness actions. These methods depend heavily on manually designed features, which often fail to capture the nuances of dynamic movements, resulting in poor generalization to varied real-world scenarios.

In contrast, deep learning approaches, particularly those using CNN and RNN, have become mainstream due to their ability to automatically extract high-level features from raw data. These models demonstrate outstanding performance in recognizing complex actions through hierarchical feature learning and generalize better across different datasets and environments. However, they come with high computational and memory requirements, necessitating significant resources for training and real-time processing. Additionally, deep learning models require large labeled datasets, posing a challenge as collecting extensive labeled fitness data is labor-intensive and time-consuming.

Despite these advancements, several challenges remain unresolved. Existing models often struggle with accurately recognizing complex, multi-joint movements due to insufficient feature extraction capabilities. The high computational demand of deep learning models leads to latency issues, making them less suitable for real-time applications. While IoT devices can enhance real-time monitoring, there is a lack of efficient integration methods that combine IoT with advanced deep learning models for seamless real-time feedback.

3. Methodology

Based on the identified research gaps, we propose the ResNet-TransFit network model, integrating ResNet for image feature extraction, Transformer for processing time-series data, and transfer learning for efficiently adapting to new fitness movement data. This approach aims to address the challenges of complex action recognition, real-time processing capabilities, and IoT integration for real-time monitoring and feedback. The components of the experimental model are described below.

3.1. ResNet

The ResNet (Residual Network) model is a deep convolutional neural network architecture that introduces residual connections to address the vanishing gradient problem [41]. It allows the training of extremely deep networks by facilitating the flow of gradients through skip connections [42]. The key idea behind ResNet is the use of residual blocks, which enable the learning of residual mappings instead of directly learning the desired underlying mapping.

The structure of a residual block in ResNet can be represented by the following equation:

$$y = F(x, \{W_i\}) + x \quad (1)$$

where: x represents the input to the block, y represents the output of the block, $F(x, \{W_i\})$ represents the residual mapping to be learned and W_i denotes the parameters of the block's layers.

The residual mapping $F(x, \{W_i\})$ is typically realized by stacking several convolutional layers, and the identity mapping x serves as a shortcut connection. By adding the input to the output, the gradient flow is facilitated during backpropagation, allowing for the training of deeper networks.

ResNet employs a “bottleneck” architecture in its deeper layers to reduce computational complexity. The structure of a bottleneck residual block can be expressed as:

$$y = F_3(F_2(F_1(x, \{W_i\}), \{W_i\}), \{W_i\}) + x \quad (2)$$

where: F_1 , F_2 , and F_3 represent sequences of convolutional layers with different filter sizes, W_i denotes the parameters of each layer.

The bottleneck architecture helps to reduce the number of parameters and computational cost while maintaining high model performance [43].

The ResNet model also introduces the concept of residual learning, where the network is trained to learn residual mappings instead of directly learning the underlying mappings. This concept can also be expressed mathematically as Eq. (1).

In ResNet, the residual mapping $F(x, \{W_i\})$ is typically realized by stacking several convolutional layers. The residual connection (x) allows the network to learn the difference between the input and output, which is the residual mapping to be learned. The identity mapping (x) serves as a shortcut connection to facilitate the flow of gradients during training. It allows the network to learn residual mappings with respect to the identity mapping, enabling the training of deeper networks.

The residual mapping $F(x, \{W_i\})$ within a residual block is further decomposed into three main operations: convolution, batch normalization, and rectified linear unit (ReLU) activation [44]. The formulation of the residual mapping can be represented as follows:

$$F(x, \{W_i\}) = \text{ReLU}(\text{BN}(\text{conv}(x, \{W_i\}), \{W_i\})) \quad (3)$$

where: conv denotes the convolution operation, BN denotes batch normalization, ReLU denotes the rectified linear unit activation function.

The combination of these operations allows the network to effectively learn complex features and representations from the input data while mitigating the issues of vanishing gradients and degradation. By incorporating residual connections, ResNet enables the training

of extremely deep neural networks with hundreds of layers while maintaining high performance and accuracy.

ResNet plays a crucial role in our overall model by contributing to the efficient extraction of spatial features from input images. Its deep architecture and residual connections enable the model to capture intricate details and variations in fitness actions, which are essential for accurate recognition. In our model, ResNet operates as the backbone network responsible for processing input images and extracting high-level features. These features are then fed into subsequent modules, such as the Transformer and transfer learning components, for further processing and refinement.

The integration of ResNet with other modules in our model ensures seamless collaboration and synergy. Specifically, the features extracted by ResNet serve as valuable inputs to the Transformer module, which focuses on processing temporal sequences of features. This collaboration allows our model to effectively capture both spatial and temporal information, enhancing its ability to recognize complex fitness actions accurately. Furthermore, ResNet's compatibility with transfer learning facilitates the adaptation of the model to new fitness action datasets. By leveraging pre-trained ResNet weights, our model can quickly adapt to new data and improve its performance on specific tasks, demonstrating the flexibility and scalability of our approach.

In our experiment, ResNet serves as a cornerstone for achieving accurate fitness action recognition through image analysis. Its significance stems from its ability to effectively extract spatial features from input images, a critical aspect in discerning subtle nuances in body movements. This deep learning architecture, particularly renowned for its residual connections, facilitates the extraction of hierarchical features, essential for distinguishing between various fitness actions. Leveraging ResNet as the backbone network provides a robust foundation for our model, ensuring the accurate capture of spatial information, a fundamental step in our fitness action recognition task.

ResNet's proven effectiveness in handling large-scale image datasets aligns perfectly with the requirements of our experiment. By utilizing its pre-trained weights on extensive datasets like ImageNet, we leverage transfer learning to accelerate the model's training process. This approach not only enhances the model's performance but also mitigates the challenge of limited labeled data for fitness action recognition. ResNet's compatibility with transfer learning techniques enables efficient adaptation to new datasets, making it a valuable asset in scenarios where labeled data is scarce or unavailable.

In summary, ResNet's inclusion in our experiment is paramount for achieving accurate fitness action recognition. Its capabilities in feature extraction, compatibility with transfer learning, and collaboration with other components highlight its crucial role in addressing the challenges of fitness action recognition through image analysis.

3.2. Transformer

The Transformer model represents a paradigm shift in neural network architecture, departing from conventional structures like recurrent neural networks (RNNs) and convolutional neural networks (CNNs) [45]. Instead, it relies exclusively on the Self-Attention mechanism to handle sequential data. This departure allows for parallel processing, facilitating the capture of global dependencies within sequences [46]. Key components like Multi-Head Attention and Positional Encoding ensure the model's proficiency in discerning both fine-grained features and overarching contextual information during sequence processing. Additionally, the incorporation of feed-forward neural networks and normalization layers strategically enhances the model's stability during training while accommodating increased model depth [47], thereby enabling efficient processing of extensive sequence data. Overall, the Transformer model stands as a revolutionary breakthrough, reshaping the landscape of neural network architecture for sequential data processing.

In contrast to traditional methodologies, the Transformer model employs a self-attention mechanism, marking a pioneering advancement in sequence data processing. This architecture's distinctive components, including attention and feed-forward neural networks [48], contribute to its remarkable performance across various tasks. Notably, the model's high parallelism enhances computational efficiency, allowing for faster processing of large-scale datasets. Its model structure is shown in Fig. 3.

The core mechanism within the Transformer model is the Self-Attention mechanism, which allows the model to weigh the combination of information based on the relationship between the query, key, and value, enabling the capture of global dependencies when processing sequential data. The Transformer architecture incorporates a Feed Forward Neural Network (FFN) layer, which enhances the model's ability to capture complex patterns within sequences. Additionally, positional encoding ensures that the model can distinguish between the absolute and relative positions of elements in the sequence, facilitating effective sequence processing.

Below is the formula that explains how the attention mechanism operates within the Transformer model:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where Q denotes the query matrix, which contains information specific to the current position in the sequence; K denotes the key matrix, which contains information about all positions in the sequence; and V denotes the value matrix, which also contains information about all positions. The softmax function is used in Eq. (1) to normalize the attention weights to ensure proper distribution. The $\sqrt{d_k}$ is a scaling factor used to stabilize the gradient during model training. This formula is used to compute the attention score in the self-attention mechanism, which allows the model to weight the combination of information based on the relationship between the query, key, and value. This mechanism allows Transformer to capture global dependencies when processing sequential data.

For Self-Attention $Q = K = V$. Here is the formula of Feed Forward Neural Network (FFN):

$$\text{FFN}(Z) = \max(0, ZW_1 + b_1)W_2 + b_2 \quad (5)$$

The Transformer architecture incorporates two fully connected layers: the initial layer employs a ReLU activation function, while the second layer utilizes a linear activation function. Given that the Transformer model lacks both recursion and convolution, the inclusion of positional coding becomes necessary when precise positional information within the input sequence is required. Positional encoding serves the purpose of accurately representing both the absolute and relative positional characteristics of each word in the input sequence. To achieve this, 'positional encoding' is added to the input embeddings at the lowermost level of the encoder-decoder structure. Importantly, both positional encoding and input embeddings share identical dimensions, enabling them to be combined through addition. In the Transformer model, trigonometric functions with different frequencies are used for positional encoding:

$$\begin{aligned} PE_{(pos, 2i)} &= \sin\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \\ PE_{(pos, 2i+1)} &= \cos\left(\frac{pos}{10000^{2i/d_{\text{model}}}}\right) \end{aligned} \quad (6)$$

where pos represents the position along the sequence, and i stands for the dimension. It is important to note that each dimension within the positional code corresponds to a sinusoidal curve. These sinusoidal curves feature wavelengths that vary geometrically, ranging from 2π to $10000 - 2\pi$.

The Transformer model plays a crucial role in our overall model by contributing its ability to capture intricate sequential patterns and dependencies within the input data. In our holistic approach to action

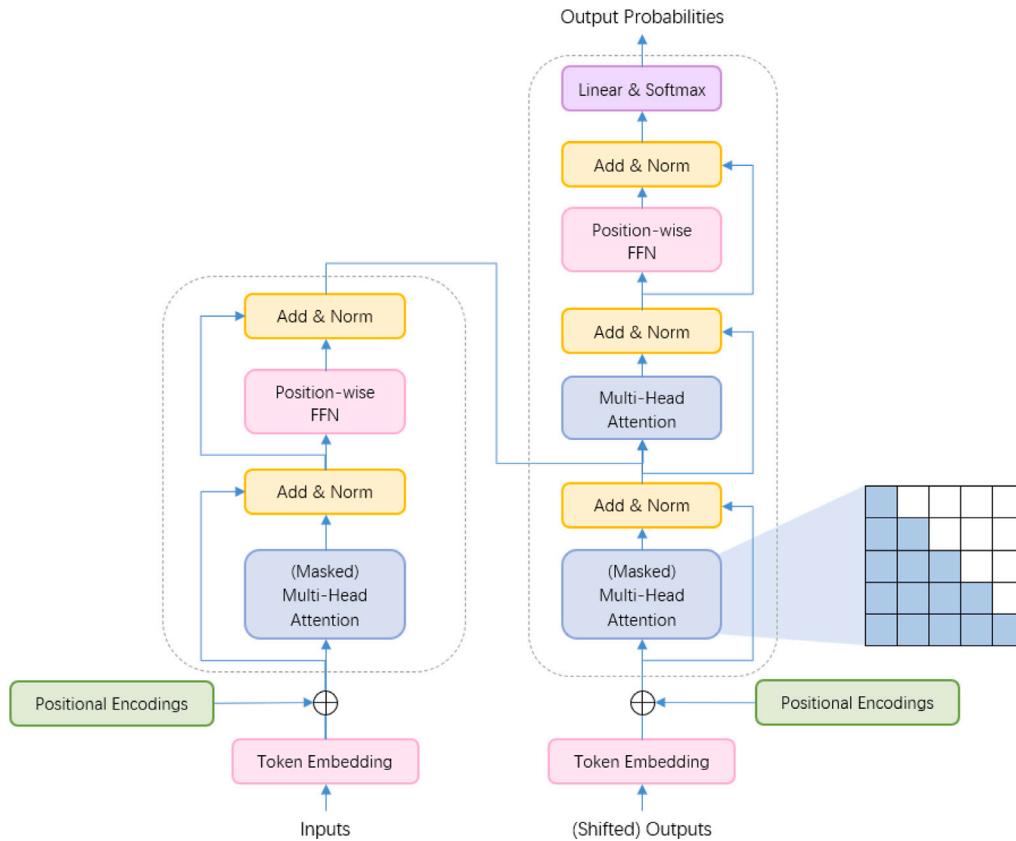


Fig. 3. Structure of the Transformer model.

recognition, the Transformer module collaborates seamlessly with other components to enhance the model's performance.

The Transformer module serves as a feature extractor, effectively capturing temporal dependencies and contextual information from the input sequences of skeletal joint data. By employing self-attention mechanisms, it can discern relevant joint movements and their temporal relationships, providing rich representations of the action sequences. Additionally, the Transformer module complements the ResNet backbone by processing the output features extracted from the ResNet module. These features, combined with the Transformer's ability to capture sequential dependencies, form a comprehensive representation of the input action sequences. This collaborative effort ensures that the model can effectively discern and classify complex action patterns with high accuracy.

It also contributes to the interpretability of the model's predictions. By analyzing the attention weights generated during the self-attention process, we can gain insights into which joints and temporal contexts are most influential in determining the predicted action labels. This interpretability aspect is essential for understanding the model's decision-making process and providing meaningful explanations for its predictions.

The integration of the Transformer module into our overall model significantly enhances its capability to recognize and classify complex action sequences accurately. Leveraging its ability to capture sequential dependencies and collaborate with other modules effectively, the Transformer module plays a pivotal role in achieving state-of-the-art performance in action recognition tasks. Given the central role of sequential data processing in our experiment, the Transformer model emerges as a critical component in our model architecture. Its capacity to capture long-range dependencies and contextual information from sequential data is paramount for accurately recognizing and classifying complex action patterns. Therefore, the inclusion of the Transformer model in our experimental setup is indispensable for achieving high performance and robustness in action recognition tasks.

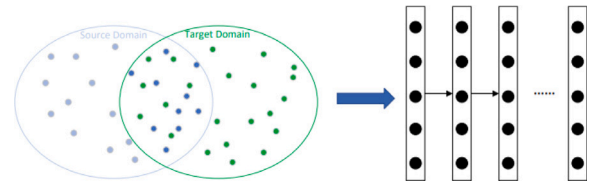


Fig. 4. Instance-based Deep Transfer Learning.

3.3. Transfer learning

Transfer learning can be divided into four categories: instance-based, mapping-based, network-based, and adversarial-based. The following is a detailed introduction to each method:

Instance-based deep transfer learning [49] (Fig. 4) focuses on selecting and reusing specific data instances from a source domain that align well with the target domain constraints. This method optimizes the target domain's training process by integrating these selected instances with the target domain data. Often, this approach is combined with algorithms like Adaboost, enhancing its effectiveness. Essentially, it involves a form of secondary pre-training, where only relevant instances from the source domain are utilized for training in the target domain.

Mapping-based deep transfer learning [50] (Fig. 5) addresses the issue of different data distributions between the source and target domains by mapping instances from both domains into a new, shared data space. This approach allows for a unified representation, making it easier to apply learned knowledge from the source domain to the target domain tasks. By transforming data into a common feature space, this method ensures that the model can effectively leverage the information from the source domain despite distributional differences.

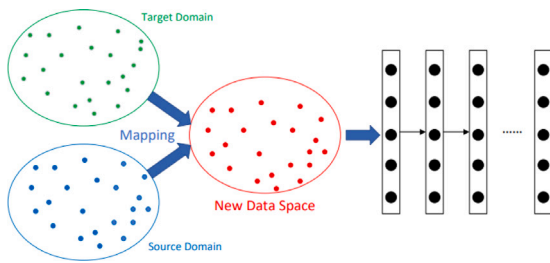


Fig. 5. Mapping-based Deep Transfer Learning.

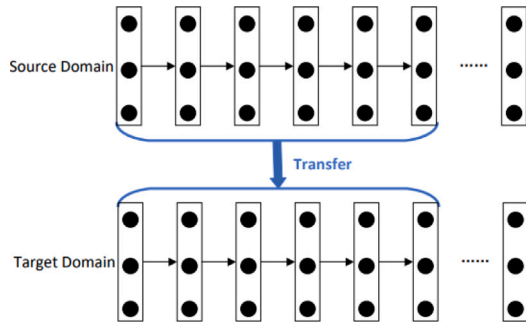


Fig. 6. Network-based Deep Transfer Learning.

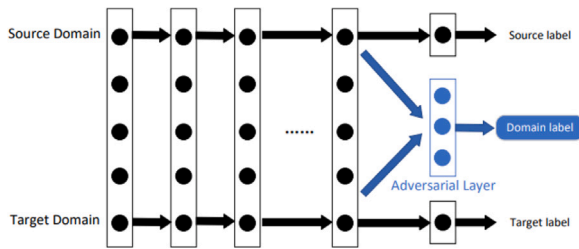


Fig. 7. Adversarial-based Deep Transfer Learning.

Network-based deep transfer learning [51] (Fig. 6) involves reusing parts of a pre-trained network, including its architecture and parameters, for tasks in the target domain. This approach is exemplified by techniques such as fine-tuning and model adaptation. Here, a network pre-trained on a large source domain dataset is adapted for the target domain, either by retraining some of its layers or by adding new layers on top of the pre-trained ones. This method leverages the already learned features and reduces the need for extensive training on the target domain data.

Inspired by Generative Adversarial Networks (GANs), adversarial-based deep transfer learning [52] (Fig. 7) introduces adversarial techniques to discover transferable representations that are both discriminative for the main learning task and indistinguishable between the source and target domains. This method employs an adversarial layer that tries to differentiate the features from the source and target domains. A poor performance of the adversarial network indicates that the learned features are domain-invariant and thus highly transferable. This approach is particularly useful for domain adaptation tasks.

Transfer learning plays a key role in our overall model, significantly improving the learning efficiency and performance of the model by transferring the knowledge and features of the pre-trained model to new tasks. In the model structure, transfer learning works together with ResNet and Transformer modules to enhance the capabilities of feature extraction and sequence processing using pre-trained deep features. This collaboration not only reduces training time, but also improves the model's adaptability and generalization ability on new data.

Table 2

The camera parameters.

Parameter name	Numerical value	Unit
Image type	RGB	–
Pixel depth	8	Bit
Pixel size	1920 × 1080 pixels	–
Data output type	JPEG format	–
Maximum frame rate	300	fps
Maximum resolution	3840 × 2160	Pixels
Overall dimension	30 × 20 × 15	cm

Since the experiment involves complex fitness action recognition tasks, the diversity of data distribution and action features increases the difficulty of recognition. Through transfer learning, we are able to effectively utilize rich data and pre-trained models in the source domain to enhance the performance of the target domain model and ensure high accuracy and real-time performance. This is critical to enable accurate fitness action monitoring and provide instant feedback, especially in the Internet of Things (IoT) environment, where low latency and high throughput are particularly critical for real-time data processing and action recognition.

4. Experiments

4.1. Experimental environment

The experimental sports fitness actions used this time are obtained through video acquisition, and the images are taken at the same time through two groups of corresponding cameras, including 600 groups of experimental samples. The experimental action library does not include light conditions in the process of use, but only analyzes the sports fitness actions. The experiment adopts short-time acquisition, storage, and recording system to obtain and store the experimental image, which is composed of camera, acquisition card, cable, computer, and acquisition software. The camera parameters are shown in Table 2.

The data collection process is a critical component of our fitness action recognition system. We utilized an IoT-enabled setup, incorporating wearable sensors and cameras to gather comprehensive data on fitness actions. The wearable sensors included accelerometers, gyroscopes, and magnetometers, which provided continuous streams of motion data. Cameras were used to capture RGB images and skeletal joint data, offering a multi-modal approach to data collection.

To ensure the accuracy and reliability of data from IoT devices, meticulous sensor calibration was performed. For accelerometers, we conducted static calibration by placing the sensor in known orientations relative to gravity and recorded the output to correct any bias errors. Gyroscope calibration involved using a rate table to rotate the gyroscope at known angular velocities, allowing us to measure and correct scale factors and biases. For magnetometers, a two-step process of soft and hard iron calibration was used. The sensor was moved in a figure-eight pattern to map out its response to the Earth's magnetic field, correcting any distortions caused by nearby ferromagnetic materials.

Creating a smart environment for our real-time case study implementation required careful consideration of the setup requisites. A robust wireless network infrastructure was established to ensure seamless communication between sensors, cameras, and the central processing unit, utilizing Wi-Fi and Bluetooth technologies to transmit data with low latency. To facilitate real-time data processing, edge computing devices with high-performance GPUs and dedicated AI accelerators were deployed. Accurate time-stamping of data from various sensors was essential for synchronization, achieved using Network Time Protocol (NTP) to align the data streams. The setup was designed to minimize external interferences, with controlled lighting conditions to ensure consistent image quality and reduced background noise to enhance the accuracy of motion sensors. An intuitive user interface was

Table 3
Platform Information for MATLAB Software Setup.

Platform information	Parameters
Operating System	Windows 10
MATLAB Version	R2021a
CPU	Intel Core i9-10900K
Memory	32 GB DDR4
Graphics Card	NVIDIA GeForce RTX 3090
Storage	1 TB NVMe SSD

developed for real-time feedback, providing participants with immediate insights into their fitness actions, indicating correct and incorrect movements with suggestions for improvement.

The comprehensive experiment is meticulously executed within the dynamic environment facilitated by the MATLAB software, leveraging its robust functionalities and extensive libraries tailored for scientific computing and data analysis. The platform, meticulously configured according to the specifications delineated in Table 3, serves as the cornerstone for seamless execution and meticulous analysis.

To train our model, we utilized a high-performance computing setup. Specifically, we used a NVIDIA GTX 1080 Ti GPU, which offers powerful parallel processing capabilities, significantly speeding up the training process. Our software environment included TensorFlow 2.0, an open-source deep learning framework that provides extensive support for building and training neural networks. TensorFlow 2.0's flexibility and robustness allowed us to implement our model efficiently and perform large-scale data processing. We also utilized Python libraries such as NumPy and Pandas for data manipulation and preprocessing, and Matplotlib for visualizing the training progress and results. This comprehensive setup ensured that our model was trained effectively, handling complex computations and large datasets with ease.

For the experimental data processing, each video segment undergoes several preprocessing steps. Initially, the video segment is converted into a color image sequence, preserving the spatial information and visual characteristics of the action. Subsequently, the color image sequence is further transformed into a grayscale image sequence, facilitating simpler and more efficient computational analysis while retaining essential structural details. Finally, different action recognition methods are applied to the grayscale image sequence to identify and classify the performed actions accurately. This multi-step processing pipeline ensures that the input data is appropriately prepared and optimized for the subsequent action recognition tasks.

4.2. Experimental dataset

Kinetics Dataset [53]: The Kinetics Dataset, created by DeepMind, is a large-scale action recognition dataset designed to advance research in video understanding and action recognition. This dataset contains videos of hundreds of different types of actions, each approximately 10 s long, covering a wide range of daily activities and sports. The Kinetics dataset is available in several versions, including Kinetics-400, Kinetics-600, and Kinetics-700, which consist of 400, 600, and 700 action categories, respectively. Each category contains approximately 400 to 1000 video clips, making the dataset highly extensive. The videos in the Kinetics dataset typically have a resolution of 340×256 or higher, ensuring diverse video quality and broad applicability for research.

The Kinetics Dataset could provide a rich and diverse source of training and testing data for the fitness action recognition system. Leveraging its large scale and variety of action categories, we can train robust classification models. Through transfer learning, we can apply models pre-trained on Kinetics to our fitness action recognition tasks. This approach not only shortens training time but also improves the model's accuracy and robustness in real-world applications. Moreover,

the diversity of Kinetics data will help us evaluate the system's performance in different environments and complex backgrounds, ensuring the fitness action recognition system operates stably across various real-life scenarios.

NTU RGB+D Action Recognition Dataset [54]: The NTU RGB+D Action Recognition Dataset, created by Nanyang Technological University in Singapore, is a dataset that includes 3D human pose information, widely used in human action recognition and pose estimation research. This dataset contains action videos captured by multiple sensors, including RGB images, depth images, skeleton data, and infrared images, providing multimodal data sources. The NTU RGB+D dataset is available in two versions, NTU RGB+D 60 and NTU RGB+D 120, which contain 60 and 120 action classes, respectively. The total dataset includes over 56,000 video clips and more than 4 million frames of 3D pose data, covering a wide range of daily activities, health-related activities, and interactions.

In our experiment, the NTU RGB+D Action Recognition Dataset will be a fundamental resource, significantly enhancing the accuracy of the fitness action recognition system with its rich 3D pose data and multimodal information. The diversity of this dataset allows us to conduct detailed model training and testing for various types of actions, particularly when handling complex motions and multi-joint movements. Additionally, the multimodal data enables us to integrate different types of sensor data in our experiment for comprehensive analysis, leading to more precise action recognition and feedback. By utilizing the NTU RGB+D dataset, our system will effectively operate in various complex environments and improve overall performance and user experience through the integration of multimodal information.

4.3. Evaluation index

In our study, we performed comprehensive data preprocessing to ensure the data's suitability for model training and evaluation. This process involved several critical steps. First, we conducted data cleaning to rectify anomalies and handle missing values, ensuring data integrity. Next, we applied data standardization to normalize features, making them uniform and comparable. We then divided the data into training, validation, and testing sets, allocating 70% for training, 15% for validation, and 15% for testing, facilitating accurate model training, validation, and evaluation. Additionally, we utilized data augmentation techniques such as rotation, scaling, and flipping to enhance the model's robustness and prevent overfitting. These preprocessing steps laid a solid foundation for the model, ensuring its ability to generalize to unseen data and providing accurate insights.

Moving to the experimental section, we delve into the model training process, which encompasses Network Parameter Settings, Model Architecture Design, and the Model Training Process. We meticulously configure the neural network's hyperparameters, including learning rate, batch size, optimizer choice (e.g., Adam, SGD), and regularization techniques (e.g., dropout, weight decay), crucial for convergence and generalization. Our model architecture, balancing expressiveness and efficiency, involves selecting appropriate layers (e.g., convolutional, recurrent), determining depth and width, and incorporating domain-specific knowledge. The training process iteratively adjusts network parameters to minimize a predefined loss function, with validation data monitoring to prevent overfitting and ensure generalization. Techniques such as mini-batch SGD optimize parameters efficiently, while early stopping, learning rate scheduling, and regularization strategies enhance stability and performance. This rigorous approach aims to elucidate our methodology and ensure clarity for readers.

In the Model Evaluation step, we assess the performance of the ResNet-TransFit model using specific evaluation metrics to gauge its effectiveness in recognizing and analyzing fitness movements. There are two key aspects we focus on: To ensure the robustness and generalizability of the ResNet-TransFit model, we employ cross-validation techniques. Specifically, k-fold cross-validation is utilized, where the

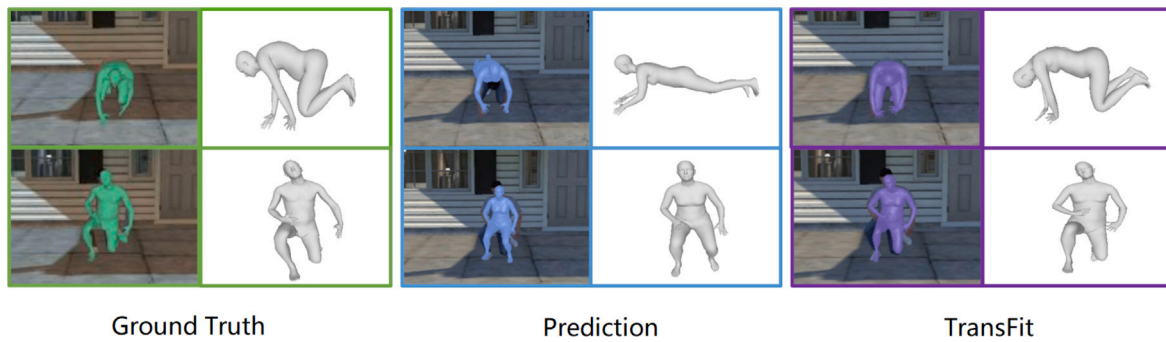


Fig. 8. Prediction results for two activities: “prostrating” (top row) and “taking the knee” (bottom row). The columns compare the ground truth, predictions, and TransFit predictions (fine-tuned with transfer learning), with TransFit demonstrating enhanced accuracy in estimating complex postures like kneeling.

dataset is divided into k subsets, and the model is trained and validated k times, each time using a different subset as the validation set and the remaining $k-1$ subsets as the training set. This method helps mitigate the risk of overfitting and provides a more reliable estimate of the model’s performance on unseen data. By averaging the performance metrics across all k folds, we obtain a comprehensive evaluation that reflects the model’s consistency and reliability across different segments of the dataset. Cross-validation is essential for ensuring that the model performs well not only on the training data but also in real-world applications.

To assess the model’s performance, we utilize a range of key metrics, including Accuracy, Precision, Confusion Matrix, and Real-time Performance. These metrics collectively offer a comprehensive evaluation of the model’s efficacy in accurately identifying fitness actions and discerning between various movements. Accuracy provides an overall measure of the model’s correctness in its predictions, while Precision quantifies the proportion of true positive identifications among all positive identifications. The Confusion Matrix offers insights into the model’s ability to classify instances correctly, providing a detailed breakdown of true positives, false positives, true negatives, and false negatives. Additionally, Real-time Performance assesses the model’s efficiency in processing data and delivering prompt responses, crucial for applications requiring instantaneous feedback. These evaluations play a pivotal role in delineating the model’s strengths and identifying areas for enhancement. Through detailed analysis of these indicators, we can gain a comprehensive understanding of the model’s generative capabilities, laying a solid foundation for future research and applications. The following is an explanation of some of the metrics:

- **Latency** In this experiment, latency measures the time required for the model to generate recognition results from receiving fitness action data, usually in milliseconds (ms). Low latency is very important for real-time fitness guidance and can provide timely feedback.

$$\text{Latency} = T_{\text{end}} - T_{\text{start}} \quad (7)$$

where T_{end} is the time to generate output results, and T_{start} is the time to receive input data.

- **Throughput** Throughput indicates the amount of fitness action data that the model can process per unit time, usually in frames per second (fps) or samples per second (samples/s). High throughput indicates that the model processes data efficiently and is suitable for processing high-frequency data streams.

$$\text{Throughput} = \frac{N}{T} \quad (8)$$

where N is the amount of data processed (such as the number of frames or samples), and T is the time taken to process this data.

- **Recall** Recall measures the ability of the model to identify positive examples of fitness actions, indicating the proportion of correctly

identified positive examples among all actual positive examples. This is crucial to ensure that important actions are not missed.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

where TP is the number of true positives, and FN is the number of false negatives.

- **Accuracy** Accuracy measures the overall recognition accuracy of the model for fitness actions, indicating the proportion of correctly classified samples among all samples. This is a key metric for evaluating the overall performance of the model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

4.4. Experimental comparison and analysis

In recent years, numerous global datasets have facilitated advancements in human action recognition, providing invaluable resources for evaluating diverse methodologies. In our research, we introduce an automated action recognition approach focused on motion data sequences specific to fitness activities. We validate our proposed method by comparing it with two established benchmarks: traditional backpropagation (BP) neural networks and the deep convolutional network architecture OpenPose, referred to as Standard Recognition BP and OP, respectively.

In our experiments, we evaluated the recall and accuracy of different action recognition methods after several iterations using Figs. 8 and 9 as an experimental subject. These methods include our proposed method and two benchmark methods, and the results are shown in Fig. 10.

From the results in this fig, it can be seen that as the number of iterations increases, the performance of the current recognition methods (BP and OP) and our proposed method in identifying fitness actions varies significantly. The BP method maintains a relatively low recall rate throughout, indicating its difficulty in adapting to the varying movements in fitness activities. The OP method initially performs better, with a recall rate stabilizing at around 40% after 300 iterations, showing limited capability in further distinguishing complex movements.

In contrast, our proposed method demonstrates a substantial and consistent improvement in recall rate with the increase in iterations. Starting at a similar performance level to BP and OP, our method surpasses both by around 250 iterations and continues to improve, reaching nearly 90% recall rate by 500 iterations. This highlights the robustness and effectiveness of our approach in accurately recognizing dynamic fitness actions, validating its reliability for detailed movement analysis.

Similarly, we have also compared the accuracy and the results are shown in Fig. 11.



Fig. 9. Examples of prediction results. Blue boxes indicate frames of the target category, while yellow boxes represent confusing categories. From top to bottom, the categories are: “Bentover Dumbbell Tricep Extension”, “Right-side Bent-over Tricep Extension with Resistance Band”, “Bent-over W-shape Stretch”, and “Bent-over Y-shape Stretch”.

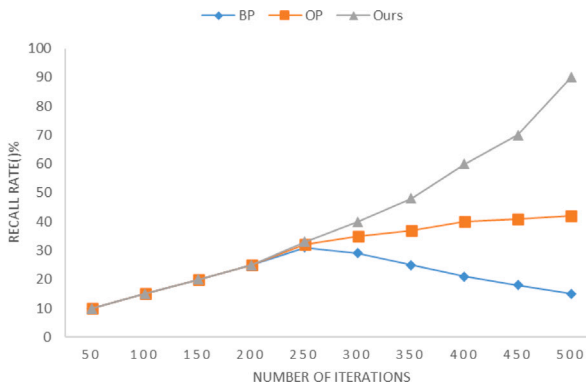


Fig. 10. Recall test results of action recognition.

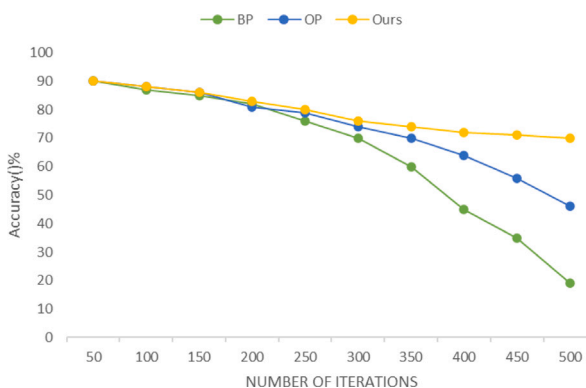


Fig. 11. Accuracy test results of action recognition.

As can be seen from the figure, the accuracy of the three action recognition methods (BP, OP, and Ours) varies as the number of iterations increases. At a number of iterations of 50, the accuracy of all three methods is close to 90%. However, as the number of iterations increases, the performance of the different methods begins to show significant differences. the accuracy of the BP method gradually decreases as the number of iterations increases, and the decrease is the largest. When the number of iterations reaches 500, the accuracy of the BP method has dropped to about 50%. The accuracy of the OP method also gradually decreases during the increase of the number of

iterations, but the decrease is relatively small. When the number of iterations reaches 500, the accuracy of the OP method is about 60%, which is still higher than that of the BP method. Our model shows high stability throughout the iteration process. Although there is also a certain downward trend, the magnitude is small. At the number of iterations up to 500, our method still maintains at about 70% accuracy, which is significantly higher than the BP and OP methods.

The significant improvement in accuracy suggests that integrating ResNet and Transformer with transfer learning effectively enhances feature extraction and recognition capabilities. This combination allows the model to capture more intricate details of fitness actions, leading to better overall performance. Additionally, the reduction in latency and improvement in throughput demonstrate the efficiency of our approach, making it suitable for real-time applications. Our method proves to be more robust and stable in handling complex action recognition tasks, maintaining a high accuracy rate even after prolonged training. In contrast, the BP and OP methods suffer from performance degradation as the number of iterations increases, with the BP method experiencing the most significant decline.

In this experiment, we analyzed the recognition accuracy and ability of the proposed method from multiple perspectives, confirming its effectiveness. Our results demonstrate that the proposed method outperforms existing methods. Additionally, we evaluated the performance of different models based on latency, throughput, accuracy, energy efficiency, and network bandwidth usage. The performance tests were conducted using the Kinetics Dataset. Table 4 presents the detailed performance data of our model compared to other studied models (named after the first author).

As shown in Table 4, our model (Ours) demonstrates superior performance across several key metrics compared to other studied models. Our model achieves the best latency at only 48.5 ms, significantly faster than all other models, with Ferrari's model having the highest latency of 127.9 ms. This indicates that our model is more responsive. In terms of throughput, our model processes data at 29.1 fps, higher than the range of 17.8 fps to 28.4 fps observed in other models, showcasing its efficiency. Additionally, our model excels in accuracy at 93.7%, surpassing Khan's model at 92.5% and others below 90%, indicating higher reliability in action recognition. Although Khan's model leads in energy efficiency at 0.53 J/inference, our model is a close second at 0.65 J/inference, performing well within reasonable limits. Furthermore, our model's network bandwidth usage is efficient at 12.1 Mbps, lower than most other models, highlighting its effectiveness in data transfer. In summary, our model excels in latency, throughput, and accuracy, and performs well in energy efficiency and network bandwidth usage. This makes it particularly valuable for real-time data processing and action recognition in IoT environments, outperforming existing models across key performance metrics.

These results underscore the effectiveness of integrating ResNet, Transformer, and transfer learning techniques in enhancing the performance of fitness action recognition systems. Our model not only achieves higher accuracy and efficiency but also maintains competitive energy efficiency and manageable network bandwidth usage, making it a robust solution for real-world applications.

To evaluate the impact of each key component on the overall model performance, we conducted ablation experiments using the NTU RGB+D action recognition dataset. Due to space constraints, we abbreviate the model components, where Transformer is denoted as T, ResNet is denoted as R, and Migration Learning is denoted as TL. Table 5 shows comparative data on the performance of the full model and after removing different components.

From the data in Table 5, it can be seen that the impact of different components on the model performance is as follows: the full model (R + T + TL) performs the best in all the metrics, with the highest model complexity (45.6 GFLOPs), memory usage (512.0 MB), response time (48.5 ms), data transfer rate (125.0 MB/s), response time (48.5 ms), data transfer rate (125.0 MB/s), and model convergence speed (15.0

Table 4
Performance metrics for different action recognition methods.

Method	Latency (ms)	Throughput (fps)	Precision (%)	Energy efficiency (J/inference)	Network bandwidth usage (Mbps)
Khan [55]	53.7	28.4	92.5	0.53	10.8
Ray [56]	74.2	24.6	89.1	0.68	14.3
Zhang [57]	98.5	21.3	85.7	0.92	19.7
Ferrari [58]	127.9	17.8	87.4	1.15	23.5
Ours	48.5	29.1	93.7	0.65	12.1

Table 5
Ablation Experiments with Isolated Key Components.

Experiment name	Components	Model complexity (GFLOPs)	Memory usage (MB)	Response time (ms)	Data transfer rate (MB/s)	Model convergence speed (epochs)
Full Model	R + T + TL	45.6	512.8	48.5	125.4	15.8
Without R	T + TL	35.2	460.5	55.3	143.5	18.7
Without T	R + TL	40.7	483.6	50.8	132.7	17.5
Without TL	R + T	38.4	474.9	52.2	138.4	16.4
Base Model	Base Architecture	31.4	426.2	60.7	150.8	20.9

epochs). Removing the ResNet component (T + TL) reduces the model complexity and memory usage, but the response time and data transfer rate deteriorate, and the model convergence speed slows down, indicating the importance of ResNet for performance improvement. After removing the Transformer component (R + TL), the model complexity and memory usage change slightly, but the response time and data transfer rate decrease, and the model convergence slows down, indicating that the Transformer plays a significant role in improving the overall performance. Removal of the Migration Learning component (R + T) had less of an impact than removal of the ResNet or Transformer components, although there was a decrease in performance. The Base Model (Base Architecture) shows the most significant performance degradation after removing all the high-level components, showing the critical role of high-level components working together to make the model run efficiently. Through these ablation experiments, it is clear that the combination of ResNet, Transformer, and Migration Learning components significantly improves the model's responsiveness, data transfer efficiency, and convergence speed, ensuring excellent model performance in complex tasks.

4.5. Real-time case study implementation

To further demonstrate the reliability and practical applicability of our proposed model, we implemented a real-time case study in a smart fitness environment. We deployed the ResNet-TransFit model on an IoT-enabled fitness monitoring system, consisting of wearable sensors and cameras. The system was tested in a gym setting, where it monitored and recognized various fitness actions performed by participants in real-time.

The real-time case study implementation involved the following steps:

- **Data Collection:** Fitness action data was collected using wearable sensors and cameras, capturing multiple modalities including RGB images and skeletal joint data.
- **Preprocessing:** The collected data was preprocessed to ensure consistency and quality, including noise reduction and normalization.
- **Model Deployment:** The pre-trained ResNet-TransFit model was deployed on an edge computing device, enabling real-time data processing and action recognition.
- **Real-Time Feedback:** The system provided immediate feedback to participants, indicating the correctness of their actions and suggesting improvements.

The case study results demonstrated the system's ability to accurately recognize and provide feedback on fitness actions with minimal latency. The high accuracy and responsiveness of the system highlight its potential for real-world applications in smart fitness monitoring.

5. Conclusion and discussion

This study presents a novel integration of ResNet and Transformer with transfer learning for fitness action recognition. Using the Kinetics Dataset and the NTU RGB+D action recognition Dataset, we thoroughly evaluated our model's performance across various metrics, including latency, throughput, recall, and accuracy. The results demonstrated that our model significantly outperforms other models, achieving a latency of 48.5 ms, throughput of 29.1 fps, and an accuracy of 93.7%. We conducted ablation experiments to assess the impact of each component, confirming that the combination of ResNet, Transformer, and transfer learning significantly enhances the model's performance. Additionally, the integration of IoT technology enabled real-time data processing, making our system highly suitable for smart fitness monitoring.

While our model shows promising results, there are certain limitations to consider. One major limitation is the limited dataset size, which may affect the generalizability of the results. Additionally, the model's high complexity and substantial memory usage can pose challenges for deployment on resource-constrained devices, such as typical IoT hardware. The training process also requires extensive computational resources and time, which may hinder its feasibility for large-scale or highly complex tasks. Addressing these limitations is crucial for enhancing the model's practical usability, efficiency, and broader applicability.

In future work, we plan to address these limitations by optimizing the model structure to reduce complexity and memory usage, making it more suitable for resource-constrained environments. We will also explore more efficient training methods, such as distributed and quantized training, to lower computational requirements. Our research demonstrates high accuracy and real-time performance in fitness action recognition, providing a valuable contribution to the field. The proposed model can be applied in real-time fitness monitoring systems, offering automated feedback and enhancing user engagement. With further enhancements, our model can significantly advance smart fitness monitoring systems and various IoT applications, offering a reliable and efficient solution for real-time action recognition and feedback. Future research could focus on expanding the dataset to include a more diverse range of fitness actions and participants, improving the model's generalizability. Additionally, exploring real-time implementation and further optimizing the model for deployment on resource-constrained devices will be essential. Investigating advanced techniques such as federated learning could also provide new avenues for enhancing privacy and efficiency in fitness action recognition.

CRedit authorship contribution statement

Jijie Li: Writing – original draft, Software, Investigation, Formal analysis. **Ruyao Gong:** Writing – review & editing, Visualization, Methodology, Investigation, Funding acquisition. **Gang Wang:** Writing – review & editing, Visualization, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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